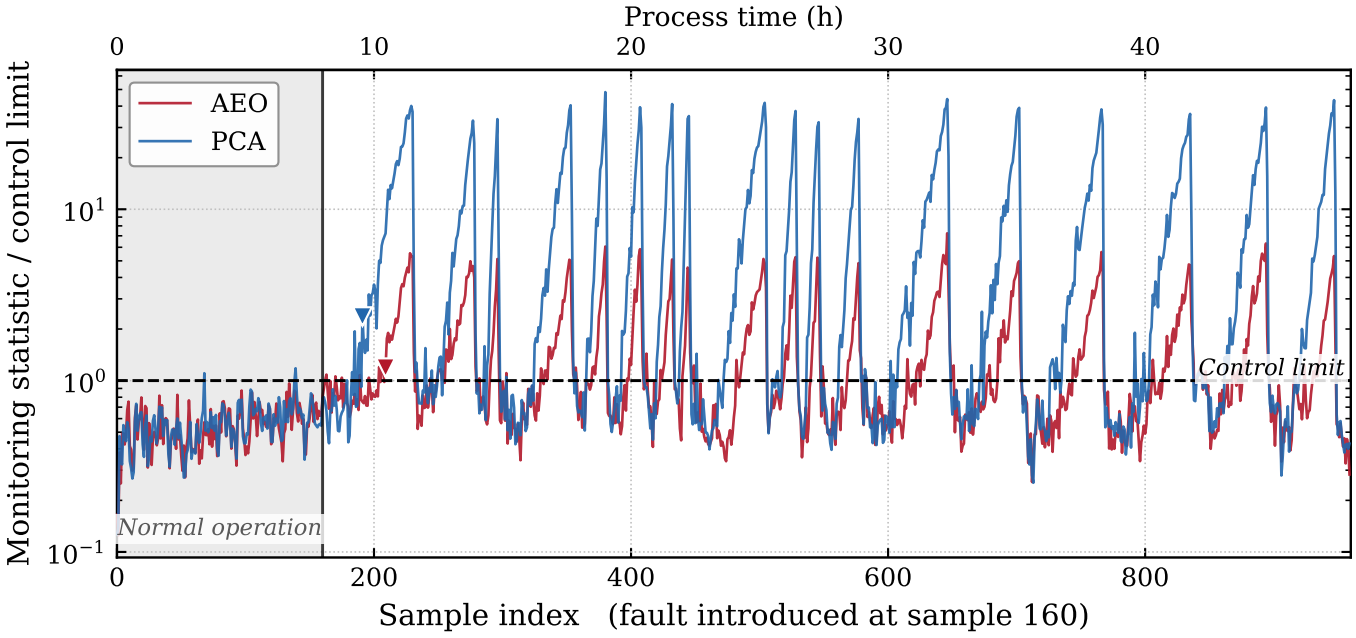
Fault Detection Performance for IDV(20)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	50.1 samples (150 min)	MDR 0.627	detected in 7/7 runs
PCA	delay	40.0 samples (120 min)	MDR 0.423	detected in 7/7 runs

PCA detects 10.1 samples (30 min) earlier; PCA missed-detection rate lower by 33%